Employee Attrition Analysis and Prediction using Data Analytics and AI

Project Report

A beginner-friendly data analytics and machine-learning project prepared for an educational internship submission.

1. Introduction

Employee attrition means employees leaving an organization. High attrition can affect productivity, recruitment costs and team stability. This project uses data analytics to identify patterns associated with attrition and a simple machine-learning model to support prediction.

2. Problem Statement

Organizations may have difficulty identifying which employee-related factors are associated with employees leaving. The project analyzes these factors and creates a simple predictive baseline.

3. Objectives

• Clean and prepare employee data.
• Perform exploratory data analysis.
• Create visualizations.
• Build a Logistic Regression classification model.
• Evaluate the model and provide recommendations.

4. Data Description

The dataset contains 1,000 synthetic employee records. Variables include age, department, job level, monthly income, years at company, job satisfaction, work-life balance, overtime, distance from home, training hours, number of companies worked and attrition.

5. Data Preprocessing

Missing numeric values were replaced with median values and categorical missing values with the most frequent category. Duplicate records were removed. Categorical variables were one-hot encoded and numeric variables were standardized.

6. Data Analysis and Visualization

The project compares attrition counts, attrition by overtime, attrition by department and monthly income across attrition groups.

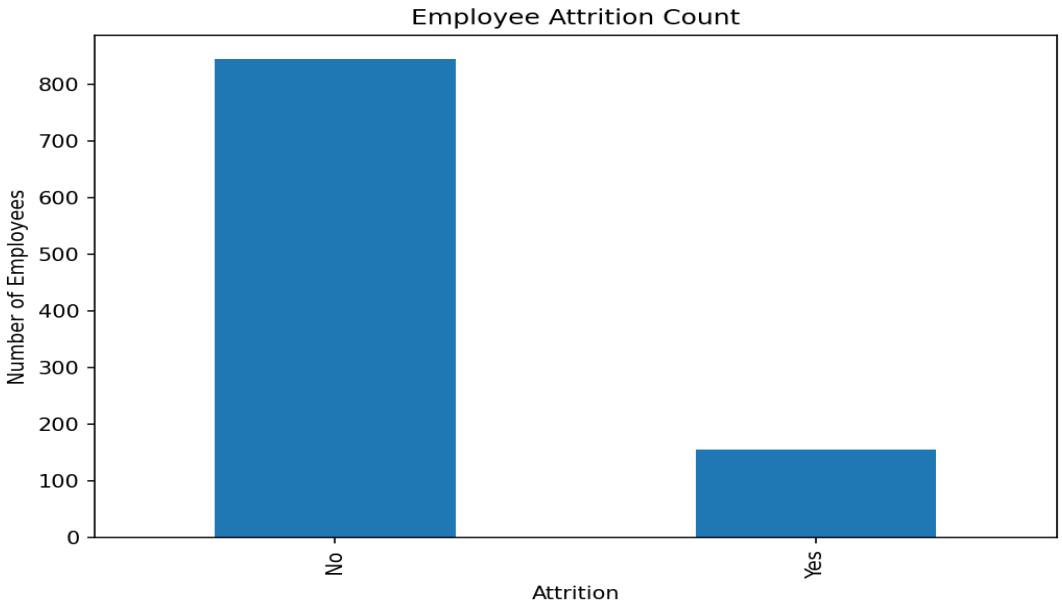


Figure 1: Employee attrition count

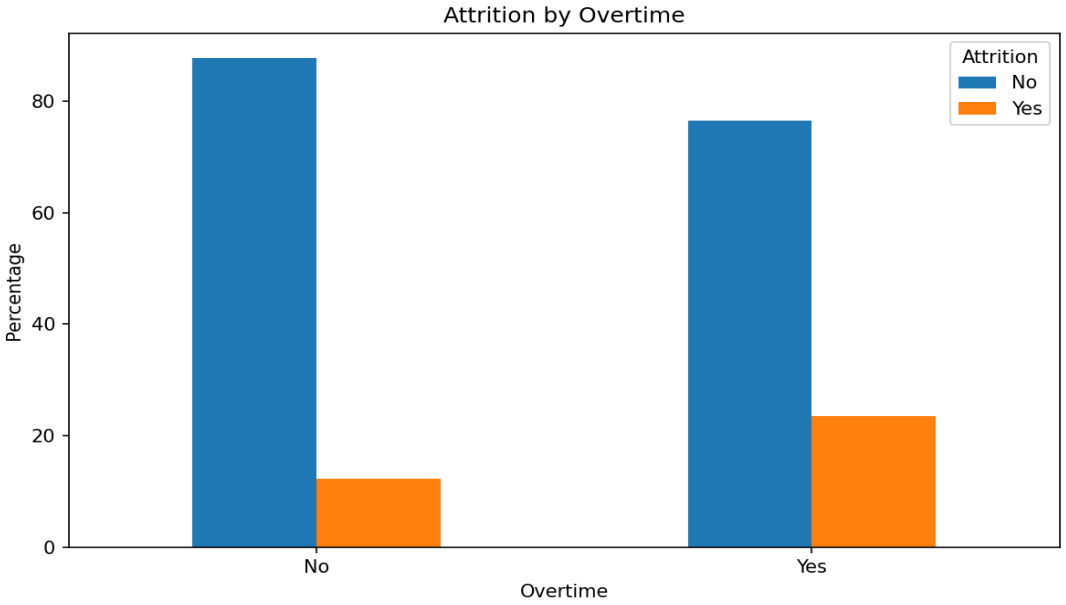


Figure 2: Attrition by overtime

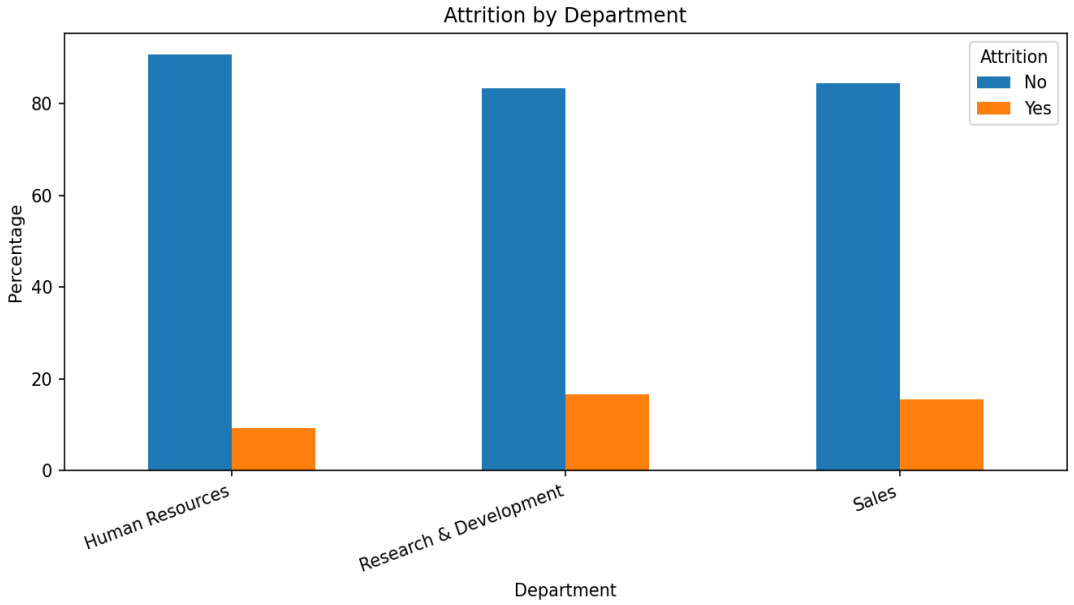


Figure 3: Attrition by department

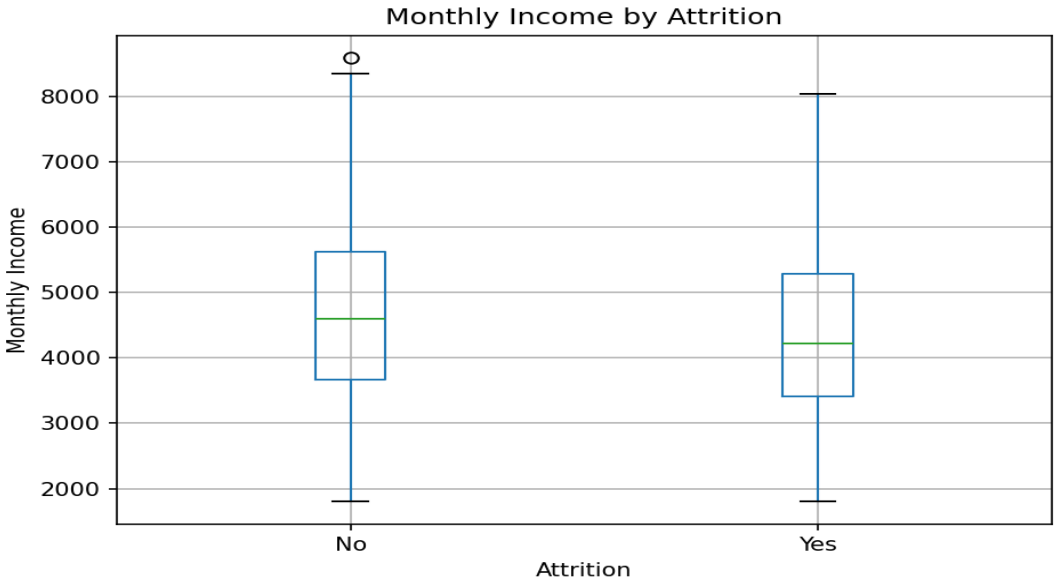


Figure 4: Monthly income by attrition

7. Machine Learning

Logistic Regression was used as a simple classification baseline. The data was split into 80% training and 20% testing sets. Class balancing was enabled because the synthetic dataset has fewer attrition cases than non-attrition cases.

Attrition rate: 15.60%
Test accuracy: 59.50%

A confusion matrix is also generated by the Python program and saved in results/charts/.

8. Key Findings

• Overtime is associated with a higher attrition rate in this synthetic dataset.
• Job satisfaction and work-life balance are useful indicators to monitor.
• Income, tenure and distance from home can help explain differences in attrition.
• Logistic Regression provides an understandable baseline for classification.

9. Recommendations

Organizations can monitor overtime, employee satisfaction and work-life balance. Regular employee feedback can help identify concerns early. Attrition predictions should be used only as a decision-support tool and not as the sole basis for employee decisions.

10. Future Scope

Future work can include a Power BI dashboard, comparison with Decision Tree and Random Forest models, additional variables, model explainability and a simple web application.

11. Limitations

The dataset is synthetic and created for educational use. Findings and model performance do not represent a real organization's workforce.

12. Tools Used

Python, Pandas, NumPy, Matplotlib and Scikit-learn.

13. References

IBM SkillsBuild Data Analytics learning resources; Scikit-learn documentation; Python, Pandas and Matplotlib documentation.